You must get familiar with the following terms.

Rows and columns: These represent the x- and y-axis of your graphs/charts.

Filter: Filters help you view a strained version of your data. For example, instead of seeing the combined sales of all the categories, you can look at a specific one, such as just *Furniture*.

Pages: Pages work on the same principle as filters, with the difference that you can actually see the changes as you shift between the paged values.

Marks: This is a visual representation of one or more rows in a data source. Mark types can be bar, line, square, and so on. You may choose to represent your data using different shapes, sizes or text.

Dimensions: These are qualitative or descriptive data, such as a name or date. These fields generally appear as column headers for rows of data, such as *Customer Name* or *Order Date*.

Measures: These are quantitative numerical data such as profit and total sales.

Start plotting

We will start by plotting a simple chart.

1. Drag *Order Date* to the *Columns* shelf, and *Sales* and *Profit* to the *Rows* shelf.
2. Tableau generates a chart that shows the total aggregated sales and profit for each year by order date. When you first create a view that includes time (in this case *Order Date*), Tableau automatically generates a line chart, as seen in Figure 3.

Another representation using the National Institutional Ranking Framework (NIRF) data set is seen in Figure 4.

Now delve a little deeper by getting the average score and count of colleges from each state (NIRF data set). Drag *State* to *Columns.* Also drag *Sheet2(Count)* to *Rows* and the *filter AVG(score).*

Drag *score* to the field 'Filters'; you will see a pop-up, 'How do you want to filter on score' which lists out many options like *#Sum*, *#Average*, *#Minimum*, *#Maximum* and so on. Select *Average*, click *Next* and finally drag *AVG(Score)* from *Filters* to *Rows.*

To gain more insight into which products drive overall sales, try adding more data. You can look at products by sub-category to see which items are the big sellers.

From the *Data* pane, drag *Category* to the *Columns* shelf and place it to the right of *YEAR(Order Date).* Drag *Sub-Category* to the *Columns* shelf. Also drag *SUM(Sales)* and *SUM(Profit)* to *Row* shelf. The bar graph depicts *Category*, *Sub-Category*, *Year of Order Date*, *Profit & Sales.* We can also see *Negative Profit* for items like *Tables* under the category *Furniture* in the year 2014

> **Note:** Tableau automatically added a colour legend and assigned a diverging colour palette, because our data includes both negative and positive values.

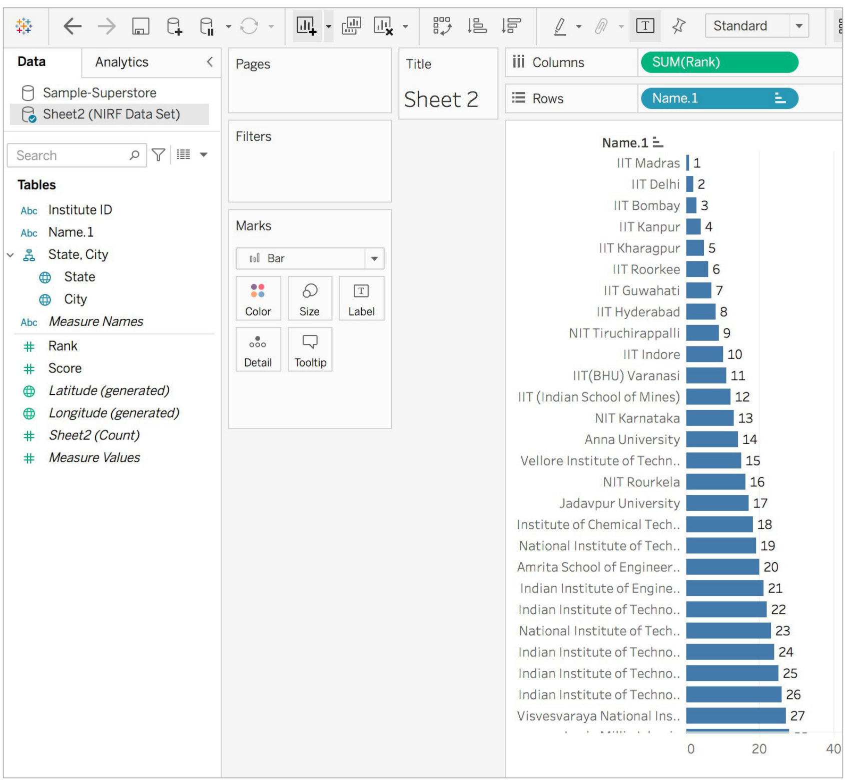

Figure 4: NIRF ranking graph

Filters

To look at the sales of a particular year or a month for a certain product, or to basically view the distinct aspects of the data, *Filters* are the way to go.

- Drag the *Dimension* to the *Filters* Shelf, select the required attribute from the dialogue box and click next.
- Let us choose *Years* and click on *Next.*
- Choose the values that you want to be a part of your *Filter.*
- Right-click on the newly generated *Filter*, and then choose *Show Filter.*
- If you feel that some of your filters can be applied to other sheets as well, then rather than repeating the steps, you can simply *Apply the Filter* to all other relevant worksheets.

How to create a calculation field

- Click on the rightmost icon near *Search – Create Calculated Field.*
- Give any name, and type your condition for colour indicator:

```
if sum(profit)>3000 THEN "Red" ELSE
"Green"
```